

CAPSTONE PROJECT CONCEPT NOTE

STD NO: 223757

Household Energy and Cost Comparison Tool for Supporting the Clean Cooking Transition in Kenya

1. Background

Access to clean cooking remains one of the most persistent development challenges in Kenya and across sub-Saharan Africa. The International Energy Agency estimates that around one billion people in sub-Saharan Africa still lack access to clean cooking, with roughly four in five households relying on polluting fuels such as firewood, charcoal, and kerosene; the resulting household air pollution contributes to an estimated 815,000 premature deaths across Africa each year (IEA, 2025; WHO, 2023). In Kenya, the Kenya National Cooking Transition Strategy identifies firewood as the primary cooking energy source for approximately 9.1 million households, the majority in rural areas (Ministry of Energy and Petroleum (MoEP, 2024a). These practices impose substantial costs on health, household budgets, women's time, and the environment, and they sit in direct tension with the national goal of universal access to clean cooking by 2028. The main barrier is cost, or how costly people believe it to be, and campaigns rarely show the money side for a household's own meals. Because cost depends heavily on which dish is cooked on which appliance, general comparisons can mislead (Perros et al., 2025). This project fills that gap by collecting primary cooking-cost data and building a simple household comparison tool.

2. Problem Statement

Most cost comparisons available to households are generic. They assume one fuel, one typical household, and a fixed set of meals, and they ignore what actually drives a household's cooking bill: household size, the dishes cooked and how often, the appliance used, and the electricity tariff band, lifeline or ordinary, which changes the unit price of power (EPRA, 2023). They also ignore that most Kenyan households use more than one stove or fuel (MoEP, 2024b). As a result, families and the field agents advising them have no simple, evidence-based way to see the real cost, savings, and payback of switching to electric cooking.

3. Main Objective

To build a tool that lets Kenyan households and clean cooking practitioners compare the energy use, monthly cost, savings, and payback period of different cooking fuels and appliances, using primary data collected through Controlled Cooking Tests.

4. Proposed Solution

The proposed solution is a Python-based decision-support tool whose distinguishing feature is that its performance engine is built directly on the Controlled Cooking Test measurements run under the MECS protocol (Leary, 2023). Its cost data will come from primary Controlled Cooking Tests measuring the energy, time, and cost of cooking seven common Kenyan dishes, beans, beef stew, rice, ugali, spinach, chapati, and chips, on ten appliances, both electric (electric pressure cooker, induction, rice cooker, air fryer, infrared, and hot plate) and non-electric (LPG, improved charcoal, kerosene, and ethanol). Generic calorific values and published efficiencies are used only as a documented fallback for fuels or appliances that fall outside this tested set.

A user will describe their household by selecting the dishes they cook and how frequently, the appliance under consideration, the household scale, and their electricity tariff band. The tool will then compute the monthly energy requirement and expenditure for each fuel option, the savings achievable by adopting electric cooking relative to a baseline stack of LPG, charcoal, and kerosene, and the resulting appliance payback period, presenting all results in shillings. Tariff-band logic is built in from the outset, so that lifeline and ordinary consumers receive estimates appropriate to their consumption level. It also supports fuel stacking, using different appliances for different dishes to match how households really cook. Prices and tariffs sit in an editable file so they can be updated, and untested dish–appliance combinations are flagged rather than guessed.

5. Proposed Methodology

The methodology is proposed in five connected phases. The first is a focused review of literature and structuring of the underlying data. Building on recent Kenyan and regional evidence on cooking energy demand, appliance efficiency, and the economics of eCooking adoption (IEA, 2025; Perros et al., 2025; Berkouwer & Dean, 2022), the CCT results will be digitised into a structured lookup table holding, for each appliance and dish, the measured energy (kWh), cooking time (minutes), and energy cost (KSh) by tariff band. Crucially, this phase documents the dataset’s missing cells and defines the tool’s behaviour when a user selects an untested combination, so that coverage gaps are managed by design rather than discovered in use. Electricity tariffs and fuel prices will be drawn from Kenya Power and the Energy and Petroleum Regulatory Authority and stored as editable parameters.

The second phase defines the tool’s architecture: the user inputs, the computational flow, and the structure of the parameter and data files. The third phase implements the calculation engine in Python, with algorithms that estimate energy requirements, convert between fuels using calorific values and efficiencies, compute monthly cooking costs by tariff band, calculate savings against the baseline stack, and derive appliance payback periods. The engine will be developed using python, with unit tests covering the core calculations so that results are reproducible and independently verifiable.

The fourth phase is validation where the tool will be validated against the collected data, benchmarked against published studies (Perros et al., 2025; Berkouwer & Dean, 2022), and tested for sensitivity to price changes, with several users’ review. The final phase prepares the technical documentation, a final report with recommendations for future improvement and deployment.

6. Deliverables

The project will deliver a working Python tool with a simple interface; a database of appliance performance, fuel prices, calorific values, and efficiencies; monthly cost comparisons across charcoal, LPG, kerosene, ethanol, and electricity; a payback calculator; a user guide and technical documentation; a final report; and recommendations for scaling affordable electric cooking in Kenya.

7. Expected Impact

By showing the clean cooking transition in the terms households care about, shillings, savings, and payback, the tool strengthens the case for electric cooking and gives electric cooking champions a practical, evidence-based aid. It supports the Kenya National eCooking Strategy (MoEP, 2024b) and Sustainable Development Goal 7 on affordable and clean energy (United Nations, 2015).

8. Implementation Plan

The project will run over nine weeks as follows:

Phase & Milestone	Key Activities	Deliverable(s)	Timeline (2026)
1. Inception & Literature Review	Review recent literature on cooking energy, appliance efficiency, and eCooking economics; confirm scope and data needs.	Literature review summary; data requirements defined.	22–27th Jun
2. Data Collection (Controlled Cooking Tests & secondary data)	Controlled Cooking Tests to measure energy, time, and cost for the selected appliances and dishes; gather EPRA fuel prices, KPLC tariffs, and calorific values; build the database.	Primary CCT dataset; structured database; editable parameter file.	29th Jun–8th Jul
3. Tool Design	Define user inputs, outputs, and the data and parameter file structure.	System architecture and model specification.	9th–14th Jul
4. Development of Calculation Engine	Build Python algorithms for energy, fuel conversion, banded monthly cost, savings, and payback; version control and unit tests.	Functional prototype and tested engine.	15th–25th July
5. Testing, Validation & Review	Check outputs against the collected data; benchmark against published studies; run price sensitivity analysis; hold one supervisor/stakeholder review.	Validation report; refined model.	26 July–3rd Aug
6. Contingency & Refinement	Buffer to absorb revisions from data collection, validation, and review.	Stabilised, validated tool.	4–8th Aug
7. Documentation & Final Reporting	Prepare technical documentation, user guide, and recommendations; compile the final report.	User guide, documentation, final report.	9–15th Aug

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